

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

a) Draw an EER diagram for the Library Management System scenario below. The system consists of several sections (fiction, children, technology) with unique section names and head librarians. Each book belongs to a section and has attributes like title, authors, ISBN, year, and publisher. A book may have multiple copies identified by access numbers. For each copy borrowed, the current borrower and due date should be tracked. Members have membership numbers, addresses, and phone numbers. Members can borrow 5 books and place hold requests on a book. Librarians have names, IDs, phones, and addresses.

The system also tracks payments made by members for course materials. A payment has a receipt number, amount, and date. Payments are classified into tuition (semester payment), examination fees, and library fines. The system should ensure data integrity and enforce constraints like unique section names, valid ISBNs, and non-negative payment amounts.

[PLACEHOLDER FIGURE] (Describe the EER diagram showing entities, attributes, and relationships)

(5)

b) Explain how you would map the Library Management System's entities and relationships to a relational model. Consider the following entities: Sections, Books, Copies, Members, Payments, and Librarians. How would you handle overlapping relationships like a member being both a student and a faculty member?

[PLACEHOLDER FIGURE] (Describe the mapping process)

(5)

c) Construct an EER diagram for the Library Management System. Ensure it includes entities, attributes, and relationships like aggregation, composition, and generalization.

[PLACEHOLDER FIGURE] (Describe the EER diagram)

(5)

d) Design a database schema for the Library Management System. Consider the entities and relationships discussed earlier. How would you normalize the data to minimize redundancy and ensure data integrity?

[PLACEHOLDER FIGURE] (Describe the database schema)

(5)

e) Explain how you would enforce constraints like unique section names, valid ISBNs, and non-negative payment amounts in the Library Management System. Consider using triggers, assertions, or other mechanisms.

[PLACEHOLDER FIGURE] (Describe the constraint enforcement mechanism) (5)

f) Map the Library Management System's binary relationships to a relational model. Consider the entities and relationships discussed earlier. How would you handle overlapping relationships like a member being both a student and a faculty member?

[PLACEHOLDER FIGURE] (Describe the mapping process) (5)

g) Design a query to retrieve all members who have borrowed books from the fiction section. Consider using joins, subqueries, or other SQL techniques.

[PLACEHOLDER FIGURE] (Describe the query) (5)

Question Q2 (20 marks)

a) Describe the relation for the University Library's book lending system, including attributes such as book title, author, publication date, and student name. (5 marks) (5)

b) Determine if the relation is in 1NF and explain why or why not. If not, propose a normalization strategy to achieve 1NF. (4 marks) (1)

c) Analyze the functional dependencies between attributes in the relation and identify any partial dependencies. Explain how these dependencies affect data integrity. (3 marks) (2)

d) Design an EER diagram for the University Library's book lending system, including entities such as 'Book', 'Student', and 'Lending'. Label the relationships between these entities. (4 marks) (4)

e) Propose a solution to address potential issues with data integrity in the book lending system, including constraints such as primary keys and foreign keys. Explain how these constraints will ensure data consistency. (2 marks) (2)

f) Explain the concept of weak entities in the context of the University Library's book lending system and provide an example of a weak entity that cannot be uniquely identified alone. (2 marks) (2)

g) Describe how the EER diagram will help ensure data consistency and integrity in the book lending system, including the role of identifying relationships and constraints. (4 marks) (4)

Question Q3 (22 marks)

- a) Describe the ER diagram for the library's book lending system, including entities such as 'Book', 'Student', and 'Librarian'. (3)
- b) Explain how to enforce referential integrity between 'Book' and 'Student' using foreign keys. (2)
- c) Design a relational schema for the 'Book' entity, including primary and secondary keys. (3)
- d) Describe the constraints on the 'Student' entity, including domain and referential integrity constraints. (4)
- e) Explain how to implement triggers in SQL to enforce business rules such as 'borrowed books cannot exceed 5'. (4)
- f) Map the ER diagram to a relational schema for the 'Librarian' entity. (3)
- g) Design an index on the 'ISBN' column of the 'Book' table to improve query performance. (3)

Question Q4 (20 marks)

- a) Describe the ER diagram for the student information system, including entities such as 'Student', 'Course', and 'Department'. Explain how to enforce referential integrity between 'Student' and 'Course'. (5 marks) (3)
- b) List three normalization rules that can be applied to the student information system. Justify why each rule is necessary. (6 marks) (3)
- c) [PLACEHOLDER FIGURE] (Describe the relation for the University Library's book lending system, including attributes such as book title, author, publication date, and student name.) (1 mark) (1)
- d) Design a relational schema for the student information system, ensuring data integrity and minimizing redundancy. (6 marks) (3)
- e) Explain how to enforce referential integrity between 'Course' and 'Department' in the relational schema designed in part d.i. (6 marks) (3)
- f) Describe the process of normalization for the student information system, including BCNF and 3NF. Justify why each form is necessary. (8 marks) (4)
- g) List two potential database management systems that can be used to implement the student information system. Compare their features and advantages. (5 marks) (3)

Question Q5 (18 marks)

- a) [PLACEHOLDER FIGURE] (Describe ER diagram for Library Management System, including entities such as 'Book', 'Student', and 'Librarian') (6)
- b) Explain how to enforce referential integrity between 'Book' and 'Student'. (2)
- c) Design a database schema for the library management system. (2)
- d) Determine if the relation is in 1NF. (3)
- e) Calculate the number of books that can be borrowed by each student. (1)
- f) Create a trigger to ensure that no book is borrowed more than once. (2)
- g) Explain how to enforce referential integrity between 'Student' and 'Librarian'. (2)